

Feasibility-Aware Quantum-Compatible Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Grid Path Planning

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Abstract—A UAV navigating in a cluttered environment must avoid obstacles while preserving line of sight to a target. Continuous formulations of this problem are commonly treated with model-predictive control, but quantum-compatible optimizers require discrete binary models. This paper introduces a feasibility-aware benchmark for a discrete occlusion-aware UAV grid-planning abstraction formulated as a higher-order unconstrained binary optimization (HUBO) problem. The binary state variable $x_{t,v}$ indicates whether the UAV occupies grid cell v at time layer t . The model combines one-hot occupancy, start, motion, obstacle, occlusion-depth, terminal-goal, and sustained near-obstacle proximity terms into a single weighted energy function. The higher-order term is not caused by the static occlusion cost; it comes from a temporal buffer-risk term that penalizes three consecutive time layers in connected near-obstacle buffer cells. In the updated 8×8 , $L = 20$ benchmark, the HUBO has 1,280 binary variables and 118,344 polynomial terms, including 4,392 cubic monomials. We compare an A* graph-search surrogate, a trajectory-level simulated annealing (SA) heuristic that evaluates the native HUBO, and D-Wave Ocean’s classical `neal` sampler after HUBO-to-QUBO reduction. A* finds a feasible target-reaching path in 4.34 ms with soft cost 257.74 and 85% visibility. Trajectory-SA obtains the best observed soft cost, 247.57, with 50/50 structural feasibility and 8/50 full-task success. `neal` obtains 6/50 structural feasibility, 4/50 goal arrival, and 0/50 full-task success, with best feasible soft cost 939.53. A tiny Qiskit QAOA experiment is included only as a QUBO compatibility check, not as a full-instance solver. The results do not claim quantum speedup; they show why feasibility-aware evaluation and explicit quadratization accounting are necessary before interpreting low-energy binary samples as useful UAV plans.

Index Terms—HUBO, QUBO, feasibility-aware benchmarking, UAV path planning, occlusion-aware planning, simulated annealing, D-Wave Ocean, QAOA, quantum-compatible optimization.

I. INTRODUCTION

Occlusion-aware UAV planning couples two requirements: the vehicle must avoid obstacles, and the target should remain visible whenever possible. Prior work has addressed related problems using continuous planning and model-predictive control (MPC), including occlusion-aware target tracking and perception-aware quadrotor control [1]–[5]. This paper studies a deliberately simplified discrete abstraction. A UAV occupies one cell of a finite grid at each time layer and must move through legal grid transitions while minimizing visibility and near-obstacle risk costs.

The goal is not to outperform classical planners on a small static grid. Instead, the goal is to formulate a reproducible binary-polynomial benchmark that can be consumed by quantum-compatible optimization workflows, including HUBO, QUBO/BQM, annealing, and reduced gate-based tests. A key lesson from the earlier version of this work was that reporting only total penalized energy is misleading: a low-energy bitstring can violate one-hot occupancy, motion, or obstacle constraints and therefore may not be a trajectory at all. We therefore center the evaluation on direct inspection of decoded trajectories.

The contributions are as follows. First, we give a complete discrete problem statement and HUBO formulation for occlusion-aware grid path planning. Second, we identify the real source of the higher-order structure: a temporal buffer-risk term, not static occlusion itself. Third, we provide a feasibility-aware evaluation protocol that reports structural feasibility, goal arrival, and full-task success separately. Fourth, we update the main instance from a horizon-tight $L = 15$ case to $L = 20$, allowing detours, target self-loops, and visibility trade-offs. Finally, we compare A*, trajectory-SA, and `neal` after HUBO-to-QUBO reduction, and we include a tiny QAOA compatibility check without claiming scalability or quantum speedup.

II. DISCRETE PLANNING MODEL

Let $G = (V, E)$ be an $N \times N$ grid graph, where V is the set of grid cells and E is the set of valid one-step transitions. In the main experiments $N = 8$, $|V| = 64$, and time layers are indexed $t = 0, \dots, L - 1$ with $L = 20$. Let $O \subset V$ be the obstacle set, $s \in V$ the start cell, and $g \in V$ the target cell. The permitted next-cell set is

$$\mathcal{N}(u) = \{u\} \cup \{v \in V : (u, v) \in E\}, \quad (1)$$

where the self-loop represents waiting or hovering.

The binary decision variable is

$$x_{t,v} \in \{0, 1\}, \quad x_{t,v} = 1 \Leftrightarrow \text{the UAV is at cell } v \text{ at time } t. \quad (2)$$

A valid binary solution decodes to a trajectory $\pi = (v_0, \dots, v_{L-1})$ only if exactly one state variable is active at every layer.

Let $\text{LOS}(u, g)$ denote the discrete Bresenham line cells strictly between u and g . The per-cell occlusion-depth surrogate is

$$C_{\text{occ}}(u) = |\text{LOS}(u, g) \cap O|. \quad (3)$$

A zero value indicates clear line of sight in this rasterized model. Let $B \subset V \setminus O$ be the set of non-obstacle cells within Chebyshev distance one of an obstacle. Cells in B are legal but risky; brief passage is allowed, while sustained residence is penalized.

III. HUBO FORMULATION

The total binary polynomial objective is

$$H(x) = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}. \quad (4)$$

The structural terms are penalty encoded:

$$H_{\text{uniq}} = \lambda_u \sum_{t=0}^{L-1} \left(\sum_{v \in V} x_{t,v} - 1 \right)^2, \quad (5)$$

$$H_{\text{start}} = \lambda_s (1 - x_{0,s}), \quad (6)$$

$$H_{\text{move}} = \lambda_m \sum_{t=0}^{L-2} \sum_{u \in V} \sum_{v \notin \mathcal{N}(u)} x_{t,u} x_{t+1,v}, \quad (7)$$

$$H_{\text{obs}} = \lambda_o \sum_{t=0}^{L-1} \sum_{v \in O} x_{t,v}. \quad (8)$$

The static-target occlusion term is linear:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t=0}^{L-1} \sum_{u \in V} C_{\text{occ}}(u) x_{t,u}. \quad (9)$$

The sole cubic term is the sustained-proximity penalty:

$$H_{\text{prox}} = \lambda_p \eta \sum_{t=1}^{L-2} \sum_{u \in B} \sum_{v \in B \cap \mathcal{N}(u)} \sum_{w \in B \cap \mathcal{N}(v)} x_{t-1,u} x_{t,v} x_{t+1,w}. \quad (10)$$

Here $\eta = 3$ in the implementation. This term penalizes a legal two-step walk that remains in connected buffer cells for three consecutive layers. Thus the model is genuinely HUBO, but the higher-order degree arises from temporal buffer-risk modeling rather than from the static occlusion-depth cost.

Goal attraction and terminal arrival are encoded as linear soft terms:

$$H_{\text{goal}} = \lambda_g \sum_{t=0}^{L-1} \sum_{v \in V} \frac{t+1}{L} \|p_v - p_g\|_2 x_{t,v}, \quad (11)$$

$$H_{\text{term}} = \lambda_{\text{term}} \sum_{v \neq g} x_{L-1,v}. \quad (12)$$

The updated instance uses $\lambda_u = \lambda_s = \lambda_m = \lambda_o = 1000$, $\lambda_{\text{occ}} = 10$, $\lambda_p = 5$, $\lambda_g = 8$, $\lambda_{\text{term}} = 50$, and $\eta = 3$.

IV. FEASIBILITY-AWARE EVALUATION

A solver output is not scored only by total HUBO energy. Instead, each binary sample is decoded and checked directly. We report: (i) structural feasibility, meaning one-hot occupancy, correct start cell, legal transitions, and no obstacle occupancy; (ii) goal arrival, meaning $v_{L-1} = g$; and (iii) full-task success, meaning structural feasibility and goal arrival for the same decoded sample. Soft cost is interpreted on feasible or full-task-successful samples, not on arbitrary bitstrings.

This distinction matters because a QUBO/BQM sampler may lower its energy by violating a structural constraint. For example, a sample can reach the target but violate motion or one-hot occupancy. Such a sample is not a valid UAV plan. Feasibility-aware benchmarking prevents these samples from being reported as successful trajectories.

V. SOLVERS AND IMPLEMENTATION

A. A* Surrogate Baseline

The A* baseline searches directly on the grid with a per-cell surrogate cost combining motion, occlusion-depth, and buffer cost [6]. It is a practical classical baseline and considers obstacle and occlusion information. However, it does not solve the exact HUBO objective because it does not represent the three-layer cubic proximity term exactly.

B. Trajectory-Level Simulated Annealing

The trajectory-level SA searches over cell sequences rather than unconstrained bitstrings. Its proposal operator keeps the start fixed and proposes each intermediate cell from the intersection of predecessor and successor neighborhoods. In the revised code, obstacle cells are removed from the candidate set; therefore the proposal operator preserves one-hot occupancy, start, motion, and obstacle feasibility by construction. SA evaluates the native HUBO energy directly, including the cubic term, but it remains a heuristic and does not certify global optimality.

C. *neal* After HUBO-to-QUBO Reduction

D-Wave Ocean's *neal* is a classical simulated-annealing sampler for binary quadratic models, not a direct HUBO solver [10]. The pipeline is therefore

$$\text{HUBO} \rightarrow \text{quadratzation} \rightarrow \text{QUBO/BQM} \rightarrow \text{neal}. \quad (13)$$

This distinction is central: *neal* searches a quadratized model with auxiliary variables and reduction penalties. Low energy in the reduced model must still be decoded and checked against the original structural constraints.

D. QAOA Scope

A Qiskit QAOA script is included only as a tiny QUBO compatibility check [9], [12]. The earlier 2×2 , $L = 2$ test had only 8 variables but could not reach the diagonal target using four-connected motion. The revised code uses $L = 3$ for a 12-variable tiny instance. This checks software compatibility with a QUBO workflow; it is not used as evidence of performance on the full 1,280-variable HUBO.

TABLE I: Updated main HUBO instance.

Quantity	Value
Grid size	8×8
Horizon L	20
Original binary variables	1,280
Total polynomial terms	118,344
Linear terms	1,280
Quadratic terms	112,672
Cubic terms	4,392
Obstacles	6
Buffer cells	24

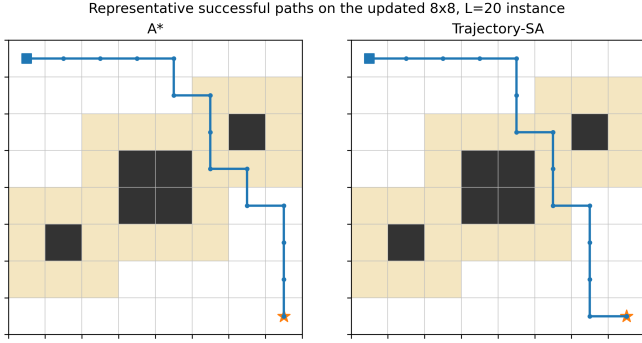


Fig. 1: Representative successful paths on the updated 8×8 , $L = 20$ instance. Both A^* and trajectory-SA reach the target and then use self-loops at the target.

VI. RESULTS

Table I summarizes the updated instance. Increasing the horizon from $L = 15$ to $L = 20$ is consequential: the Manhattan distance from $(0,0)$ to $(7,7)$ is 14, so the previous $L = 15$ case allowed exactly 14 moves and forced shortest monotone paths. The updated horizon gives additional layers for detours or target self-loops.

The A^* baseline finds a feasible, target-reaching path with 85% visibility in 4.34 ms. Trajectory-SA finds a slightly lower best feasible soft cost, 247.57, but requires substantially more computation and reaches the target in only 8 of 50 runs. *neal*, which operates on the reduced QUBO rather than on the native HUBO, obtains 6/50 structural feasibility and 4/50 goal arrival; however, none of the 50 *neal* samples is both structurally feasible and target-reaching. Its best structurally feasible soft cost is 939.53.

The penalty-weight sweep for *neal* shows the expected pattern: weak structural penalties permit low-energy but infeasible samples, while larger penalties improve feasibility. In the updated run, structural feasibility was 0/15 for $\lambda \in \{50, 100, 300, 600\}$, 4/15 at $\lambda = 1000$, and 5/15 at $\lambda = 2000$. The best feasible soft cost improved from 915.4 to 590.6 between $\lambda = 1000$ and $\lambda = 2000$, but still remained worse than A^* and trajectory-SA.

A. Tiny QAOA Compatibility Check

The revised QAOA test uses a 2×2 , $L = 3$ QUBO with 12 variables and 38 terms. Classical SA and *neal* both find

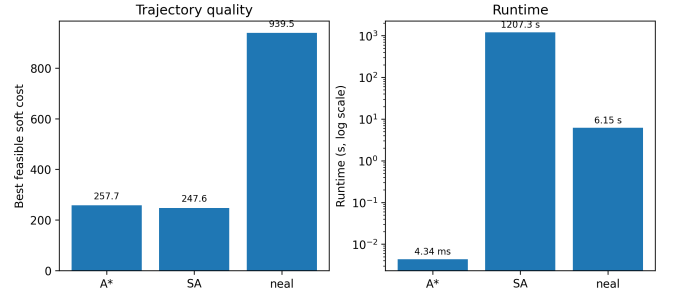


Fig. 2: Best feasible soft cost and runtime from the updated benchmark. SA obtains the best observed soft cost, A^* is much faster, and *neal* struggles after quadratization. Runtime is plotted on a log scale.

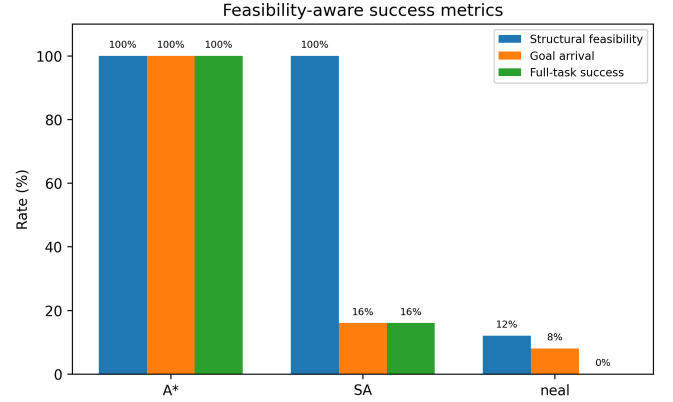


Fig. 3: Feasibility-aware success metrics. *neal* has nonzero structural feasibility and goal arrival separately, but zero full-task success in this run.

feasible target-reaching samples with energy -195.45 . Qiskit QAOA with one layer and 25 SPSA iterations returns a target-reaching sample with energy -148.11 , but the decoded trajectory has one motion violation and is therefore not structurally feasible. This reinforces the role of QAOA here: it is a small software-compatibility check, not a performance claim for the full HUBO benchmark.

VII. DISCUSSION

The updated results support three conclusions. First, the $L = 20$ instance is a more meaningful planning benchmark than the previous horizon-tight case because detours and target self-loops are possible. Second, A^* and trajectory-SA provide strong classical baselines. A^* is fast and target-reaching, while trajectory-SA obtains the lowest observed soft cost but is slow and unreliable in goal arrival. Third, *neal* on the quadratized QUBO has difficulty producing samples that are both structurally feasible and target-reaching under the tested settings.

These results should not be interpreted as evidence against quantum optimization in general. They show that a penalty-encoded binary formulation can be difficult for an unconstrained BQM sampler, especially after quadratization. They

TABLE II: Updated benchmark results. Full-task success means structural feasibility and goal arrival in the same decoded sample. Visibility for stochastic methods is reported as mean over all samples and, when available, mean over full-task-successful samples.

Metric	A*	Trajectory-SA	neal on quadratized QUBO
Samples	1 run	50 runs	50 reads
Runtime	4.34 ms	1207.26 s	6.15 s
Best feasible soft cost	257.74	247.57	939.53
Structural feasibility	1/1	50/50	6/50
Goal arrival	1/1	8/50	4/50
Full-task success	1/1	8/50	0/50
Visibility, all samples	85.0%	77.5%	84.4%
Visibility, full successes	85.0%	80.6%	N/A

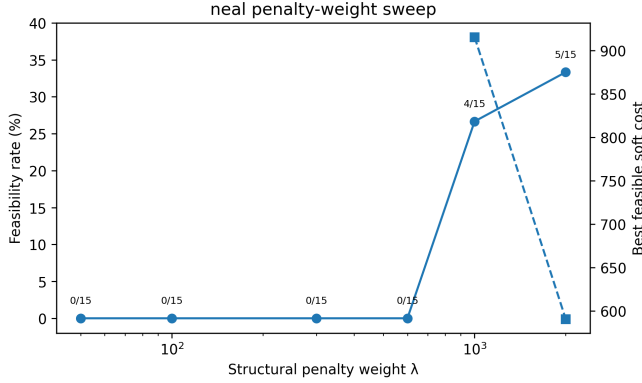


Fig. 4: Penalty-weight sweep for `neal` on the quadratized QUBO. Feasibility improves only after structural penalties become large relative to soft trajectory costs.

also show why feasibility-aware evaluation is necessary: reporting only best energy can overstate solver performance if low-energy samples are not valid trajectories.

VIII. LIMITATIONS

The model remains a grid abstraction. It has no quadrotor dynamics, heading, velocity, acceleration, camera field-of-view dynamics, or continuous controls. The target is static, so the occlusion term is linear; a moving-target formulation with target decision variables would require a separate derivation. QAOA is included only as a tiny compatibility check and is not a scalable performance claim. Finally, A* is a graph-search surrogate rather than an exact optimizer of the native HUBO polynomial. A future exact CP-SAT or MILP formulation would be useful as a global reference for the weighted objective.

IX. CONCLUSION

This paper presents a feasibility-aware HUBO benchmark for occlusion-aware UAV grid path planning. The higher-order structure comes from sustained temporal proximity to obstacle-buffer cells, while static occlusion is modeled as a line-of-sight depth cost. The updated 8×8 , $L = 20$ instance contains 1,280 variables and 4,392 cubic monomials. A* and trajectory-level SA are strong classical baselines; `neal` after

quadratization produces fewer feasible and goal-reaching samples. The main contribution is therefore not a quantum speedup claim, but a reproducible quantum-compatible benchmark and an evaluation protocol that prevents infeasible low-energy samples from being misreported as successful UAV paths.

CODE AVAILABILITY

The code package used for this revision includes the HUBO builder, A* baseline, trajectory-SA, `neal` benchmark, QAOA tiny compatibility check, plotting scripts, and result logs. A public repository or supplementary-material link should be inserted before submission.

REFERENCES

- [1] J. Chen, T. Liu, and S. Shen, “Tracking a moving target in cluttered environments using a quadrotor,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2016, pp. 446–453.
- [2] K. Hausman, G. Kahn, S. Patil, J. Muller, K. Goldberg, P. Abbeel, and G. S. Sukhatme, “Occlusion-aware multi-robot 3D tracking,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2016, pp. 1863–1870.
- [3] B. Penin, P. Robuffo Giordano, and F. Chaumette, “Vision-based reactive planning for aggressive target tracking while avoiding collisions and occlusions,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3725–3732, 2018.
- [4] D. Falanga, P. Foehn, P. Lu, and D. Scaramuzza, “PAMPC: Perception-aware model predictive control for quadrotors,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2018.
- [5] H. Masnavi, V. K. Adajania, K. Kruusamäe, and A. K. Singh, “Real-time multi-convex model predictive control for occlusion-free target tracking with quadrotors,” *IEEE Access*, vol. 10, pp. 29009–29031, 2022.
- [6] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Trans. Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
- [7] I. G. Rosenberg, “Reduction of bivalent maximization to the quadratic case,” *Cahiers du Centre d’Etudes de Recherche Operationnelle*, vol. 17, pp. 71–74, 1975.
- [8] F. Glover, G. Kochenberger, and Y. Du, “Quantum bridge analytics I: A tutorial on formulating and using QUBO models,” *4OR*, vol. 17, pp. 335–371, 2019.
- [9] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” arXiv:1411.4028, 2014.
- [10] D-Wave Systems, “dwave-neal simulated annealing sampler,” software package.
- [11] D-Wave Systems, “dimod: A shared API for binary quadratic models,” software package.
- [12] Qiskit contributors, “Qiskit, Qiskit Optimization, Qiskit Algorithms, and Qiskit Aer,” software packages.